HRMS

10 訓練管理 — API 規格書

HR10 TRAINING Service — API Specification

系統名稱	版本
人力資源暨專案管理系統	**1.0**
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目錄

HR10 訓練管理服務 API 詳細規格

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1. API 總覽

1.1 端點清單

#	端點	方法	說明	Controller
1	`/api/v1/training/courses`	POST	建立課程	HR10CourseCmdController
2	`/api/v1/training/courses`	GET	查詢課程列表	HR10CourseQryController
3	`/api/v1/training/courses/{id}`	GET	查詢課程詳情	HR10CourseQryController
4	`/api/v1/training/courses/{id}`	PUT	更新課程	HR10CourseCmdController
5	`/api/v1/training/courses/{id}/publish`	PUT	發布課程	HR10CourseCmdController
6	`/api/v1/training/courses/{id}/close`	PUT	關閉報名	HR10CourseCmdController
7	`/api/v1/training/courses/{id}/complete`	PUT	完成課程	HR10CourseCmdController
8	`/api/v1/training/enrollments`	POST	報名課程	HR10EnrollmentCmdController
9	`/api/v1/training/enrollments`	GET	查詢報名列表	HR10EnrollmentQryController
10	`/api/v1/training/enrollments/{id}/approve`	PUT	審核通過	HR10EnrollmentCmdController
11	`/api/v1/training/enrollments/{id}/reject`	PUT	審核拒絕	HR10EnrollmentCmdController
12	`/api/v1/training/enrollments/{id}/cancel`	PUT	取消報名	HR10EnrollmentCmdController

#	端點	方法	說明	Controller
13	`/api/v1/training/enrollments/{id}/attendance`	PUT	確認出席	HR10EnrollmentCmdController
14	`/api/v1/training/enrollments/{id}/complete`	PUT	完成訓練	HR10EnrollmentCmdController
15	`/api/v1/training/my`	GET	我的訓練	HR10EnrollmentQryController
16	`/api/v1/training/certificates`	POST	新增證照	HR10CertificateCmdController
17	`/api/v1/training/certificates`	GET	查詢證照列表	HR10CertificateQryController
18	`/api/v1/training/certificates/{id}`	GET	查詢證照詳情	HR10CertificateQryController
19	`/api/v1/training/certificates/{id}`	PUT	更新證照	HR10CertificateCmdController
20	`/api/v1/training/certificates/{id}`	DELETE	刪除證照	HR10CertificateCmdController
21	`/api/v1/training/certificates/expiring`	GET	即將到期證照	HR10CertificateQryController
22	`/api/v1/training/my/hours`	GET	我的訓練時數	HR10ReportQryController
23	`/api/v1/training/statistics`	GET	訓練統計報表	HR10ReportQryController
24	`/api/v1/training/statistics/export`	GET	匯出統計報表	HR10ReportQryController

1.2 報名狀態流程

```
REGISTERED (已報名)
     ↓
  ┌──┴──┐
  ↓     ↓
APPROVED  REJECTED (已拒絕)
(已核准)
  ↓
ATTENDED (已出席)
  ↓
  ├── COMPLETED (已完成)
  └── NO_SHOW (未到)
```

1.3 課程狀態流程

```
DRAFT (草稿) → OPEN (開放報名) → CLOSED (報名截止) → COMPLETED (已結束)
                                          ↓
                                   CANCELLED (已取消)
```

1.4 通用 Headers

Header	必填	說明
`Authorization`	✅	Bearer Token
`X-Tenant-Id`	✅	租戶識別碼
`Content-Type`	✅	application/json

2. 課程管理 API

2.1 建立課程

基本資訊

項目	內容
端點	`POST /api/v1/training/courses`
方法	POST
Controller	HR10CourseCmdController
Service	CreateCourseServiceImpl
權限	`training:course:create`

用途說明

項目	說明
業務場景	HR 或訓練單位建立新的訓練課程
使用者	HR 人員、訓練管理員
前置條件	已取得年度訓練計畫核准

業務邏輯

```
1. 驗證規則
   ├── 課程名稱必填且不超過 255 字元
   ├── 課程代碼唯一
   ├── 訓練時數 > 0
   ├── 開始日期 <= 結束日期
   └── 最大人數 > 0（若設定）

2. 處理步驟
   ├── 產生課程代碼（若未指定）
   ├── 建立課程記錄，狀態為 DRAFT
   └── 發布 CourseCreatedEvent

3. 課程代碼規則
   └── TRN-{YYYYMM}-{序號} (e.g., TRN-202512-001)
```

Request Body

```
{
  "courseCode": "TRN-202512-001",
  "courseName": "React 進階開發實戰",
  "courseType": "INTERNAL",
  "deliveryMode": "OFFLINE",
  "category": "TECHNICAL",
  "description": "本課程將深入介紹 React Hooks、Redux Toolkit 等進階技術...",
  "instructor": "李講師",
  "instructorInfo": {
    "name": "李講師",
    "title": "資深前端工程師",
    "company": "內部講師"
  },
  "durationHours": 8,
  "maxParticipants": 30,
  "minParticipants": 10,
  "startDate": "2025-12-15",
  "endDate": "2025-12-15",
  "startTime": "09:00",
  "endTime": "18:00",
  "location": "台北辦公室 5F 訓練教室",
  "cost": 0,
  "isMandatory": false,
  "targetAudience": ["RD", "QA"],
  "prerequisites": "具備 React 基礎知識",
  "enrollmentDeadline": "2025-12-10"
}
```

欄位	類型	必填	驗證規則	說明
`courseCode`	String	❌	唯一，最大 50 字	課程代碼（可自動產生）
`courseName`	String	✅	長度 1-255	課程名稱
`courseType`	Enum	✅	INTERNAL/EXTERNAL	課程類型
`deliveryMode`	Enum	✅	ONLINE/OFFLINE/HYBRID	授課方式
`category`	Enum	❌	見類別列表	課程類別
`description`	String	❌	最大 2000 字	課程說明
`instructor`	String	❌	最大 100 字	講師姓名
`instructorInfo`	Object	❌	-	講師詳細資訊
`durationHours`	Decimal	✅	> 0	訓練時數
`maxParticipants`	Integer	❌	> 0	最大人數
`minParticipants`	Integer	❌	> 0	最小開課人數
`startDate`	Date	✅	不早於今天	開始日期
`endDate`	Date	✅	>= startDate	結束日期
`startTime`	Time	❌	HH:mm	開始時間
`endTime`	Time	❌	HH:mm	結束時間
`location`	String	❌	最大 255 字	上課地點
`cost`	Decimal	❌	>= 0	課程費用
`isMandatory`	Boolean	❌	-	是否必修
`targetAudience`	Array	❌	-	目標對象（部門代碼）
`prerequisites`	String	❌	最大 500 字	先修條件
`enrollmentDeadline`	Date	❌	< startDate	報名截止日

課程類型 (courseType):

- `INTERNAL` - 內訓（公司內部講師）
- `EXTERNAL` - 外訓（外部機構）

授課方式 (deliveryMode):

- `ONLINE` - 線上課程
- `OFFLINE` - 實體課程
- `HYBRID` - 混合式

課程類別 (category):

- `TECHNICAL` - 技術類
- `MANAGEMENT` - 管理類
- `SOFT_SKILL` - 軟技能
- `COMPLIANCE` - 法規遵循
- `ORIENTATION` - 新人訓練
- `SAFETY` - 安全衛生
- `OTHER` - 其他

Response Body

成功回應 (201 Created)

```
{
  "code": "SUCCESS",
  "message": "課程建立成功",
  "data": {
    "courseId": "course-uuid-001",
    "courseCode": "TRN-202512-001",
    "courseName": "React 進階開發實戰",
    "status": "DRAFT",
    "createdAt": "2025-12-01T10:00:00Z"
  }
}
```

錯誤碼

HTTP 狀態碼	錯誤碼	說明	處理建議
400	`TRN_COURSE_NAME_REQUIRED`	課程名稱必填	請填寫課程名稱
400	`TRN_COURSE_CODE_EXISTS`	課程代碼已存在	請使用其他代碼
400	`TRN_INVALID_DATE_RANGE`	日期範圍無效	結束日期需 >= 開始日期
400	`TRN_INVALID_HOURS`	訓練時數無效	時數需 > 0

領域事件

CourseCreatedEvent

```
{
  "eventType": "CourseCreated",
  "aggregateId": "course-uuid-001",
  "aggregateType": "TrainingCourse",
  "payload": {
    "courseId": "course-uuid-001",
    "courseCode": "TRN-202512-001",
    "courseName": "React 進階開發實戰",
    "courseType": "INTERNAL",
    "durationHours": 8,
    "startDate": "2025-12-15",
    "createdBy": "user-uuid"
  }
}
```

Topic	訂閱服務
`training.course.created`	-

2.2 查詢課程列表

基本資訊

項目	內容
端點	`GET /api/v1/training/courses`
方法	GET
Controller	HR10CourseQryController
Service	GetCoursesServiceImpl
權限	`training:course:read`

Query Parameters

參數	類型	必填	說明
`status`	Enum	❌	課程狀態篩選
`courseType`	Enum	❌	課程類型
`category`	Enum	❌	課程類別
`keyword`	String	❌	關鍵字搜尋（名稱、代碼）
`dateFrom`	Date	❌	開始日期起
`dateTo`	Date	❌	開始日期迄
`enrollable`	Boolean	❌	僅顯示可報名課程
`page`	Integer	❌	頁碼（預設 1）
`size`	Integer	❌	每頁筆數（預設 20）

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "data": {
    "items": [
      {
        "courseId": "course-uuid-001",
        "courseCode": "TRN-202512-001",
        "courseName": "React 進階開發實戰",
        "courseType": "INTERNAL",
        "courseTypeLabel": "內訓",
        "deliveryMode": "OFFLINE",
        "deliveryModeLabel": "實體課程",
        "category": "TECHNICAL",
        "categoryLabel": "技術類",
        "instructor": "李講師",
        "durationHours": 8,
        "maxParticipants": 30,
        "currentEnrollments": 15,
        "availableSeats": 15,
        "startDate": "2025-12-15",
        "endDate": "2025-12-15",
        "location": "台北辦公室 5F",
        "status": "OPEN",
        "statusLabel": "報名中",
        "isMandatory": false,
        "enrollmentDeadline": "2025-12-10",
        "canEnroll": true
      }
    ],
    "pagination": {
      "page": 1,
      "size": 20,
      "totalItems": 25,
      "totalPages": 2
    }
  }
}
```

2.3 查詢課程詳情

基本資訊

項目	內容
端點	`GET /api/v1/training/courses/{id}`
方法	GET
Controller	HR10CourseQryController
Service	GetCourseDetailServiceImpl
權限	`training:course:read`

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "data": {
    "courseId": "course-uuid-001",
    "courseCode": "TRN-202512-001",
    "courseName": "React 進階開發實戰",
    "courseType": "INTERNAL",
    "courseTypeLabel": "內訓",
    "deliveryMode": "OFFLINE",
    "deliveryModeLabel": "實體課程",
    "category": "TECHNICAL",
    "categoryLabel": "技術類",
    "description": "本課程將深入介紹...",
    "instructorInfo": {
      "name": "李講師",
      "title": "資深前端工程師",
      "company": "內部講師"
    },
    "durationHours": 8,
    "maxParticipants": 30,
    "minParticipants": 10,
    "currentEnrollments": 15,
    "startDate": "2025-12-15",
    "endDate": "2025-12-15",
    "startTime": "09:00",
    "endTime": "18:00",
    "location": "台北辦公室 5F 訓練教室",
    "cost": 0,
    "isMandatory": false,
    "targetAudience": ["RD", "QA"],
    "prerequisites": "具備 React 基礎知識",
    "enrollmentDeadline": "2025-12-10",
    "status": "OPEN",
    "statusLabel": "報名中",
    "enrollmentStats": {
      "registered": 5,
      "approved": 10,
      "rejected": 2,
      "cancelled": 1
    },
    "myEnrollment": {
      "enrollmentId": "enroll-uuid-001",
      "status": "APPROVED",
      "enrolledAt": "2025-12-02T14:00:00Z"
    },
    "createdBy": "user-uuid",
    "createdByName": "HR Admin",
    "createdAt": "2025-12-01T10:00:00Z",
    "updatedAt": "2025-12-02T09:00:00Z"
```

```
  }
}
```

2.4 更新課程

基本資訊

項目	內容
端點	`PUT /api/v1/training/courses/{id}`
方法	PUT
Controller	HR10CourseCmdController
Service	UpdateCourseServiceImpl
權限	`training:course:update`

業務邏輯

```
1. 驗證規則
   ├── 課程必須存在
   ├── 只有 DRAFT 或 OPEN 狀態可編輯
   └── 若已有報名，部分欄位受限制

2. 受限制欄位（已有報名時）
   ├── 開始日期僅可延後
   ├── 最大人數不可少於目前報名人數
   └── 課程時數不可減少
```

Request Body

```
{
  "courseName": "React 進階開發實戰（更新版）",
  "description": "更新課程說明...",
  "maxParticipants": 35,
  "location": "台北辦公室 6F 大會議室"
}
```

Response Body

成功回應 (200 OK)

```
{
  "code": "SUCCESS",
  "message": "課程更新成功",
  "data": {
    "courseId": "course-uuid-001",
    "courseName": "React 進階開發實戰（更新版）",
    "updatedAt": "2025-12-03T10:00:00Z"
  }
}
```

2.5 發布課程

基本資訊

項目	內容
端點	`PUT /api/v1/training/courses/{id}/publish`
方法	PUT
Controller	HR10CourseCmdController
Service	PublishCourseServiceImpl
權限	`training:course:publish`

用途說明

項目	說明
業務場景	將草稿課程發布為開放報名狀態
使用者	HR 人員、訓練管理員
後續動作	員工可開始報名

業務邏輯

```
1. 驗證規則
   ├── 課程必須存在
   ├── 課程狀態必須為 DRAFT
   ├── 必填欄位完整（日期、時數）
   └── 開始日期需在未來

2. 處理步驟
   ├── 更新課程狀態為 OPEN
   ├── 發布 CoursePublishedEvent
   └── 若為必修，發送通知給目標對象
```

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "message": "課程已發布，開放報名",
  "data": {
    "courseId": "course-uuid-001",
    "status": "OPEN",
    "publishedAt": "2025-12-03T10:00:00Z"
  }
}
```

領域事件

CoursePublishedEvent

```json
{
  "eventType": "CoursePublished",
  "aggregateId": "course-uuid-001",
  "payload": {
    "courseId": "course-uuid-001",
    "courseName": "React 進階開發實戰",
    "isMandatory": false,
    "targetAudience": ["RD", "QA"],
    "startDate": "2025-12-15",
    "enrollmentDeadline": "2025-12-10"
  }
}
```

Topic	訂閱服務
`training.course.published`	Notification（若為必修）

2.6 關閉報名

基本資訊

項目	內容
端點	`PUT /api/v1/training/courses/{id}/close`
方法	PUT
Controller	HR10CourseCmdController
Service	CloseCourseEnrollmentServiceImpl
權限	`training:course:close`

業務邏輯

```
1. 驗證規則
   ├── 課程必須存在
   └── 課程狀態必須為 OPEN

2. 處理步驟
   ├── 更新課程狀態為 CLOSED
   ├── 拒絕所有待審核的報名
   └── 發布 CourseClosedEvent
```

Request Body

```json
{
  "reason": "報名截止"
}
```

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "message": "課程已關閉報名",
  "data": {
    "courseId": "course-uuid-001",
    "status": "CLOSED",
    "closedAt": "2025-12-10T18:00:00Z"
  }
}
```

2.7 完成課程

基本資訊

項目	內容
端點	`PUT /api/v1/training/courses/{id}/complete`
方法	PUT
Controller	HR10CourseCmdController
Service	CompleteCourseServiceImpl
權限	`training:course:complete`

用途說明

項目	說明
業務場景	課程結束後，結算訓練結果
使用者	HR 人員、訓練管理員
後續動作	更新學員訓練時數

業務邏輯

```
1. 驗證規則
   ├── 課程必須存在
   ├── 課程狀態必須為 CLOSED
   └── 課程結束日期已過

2. 處理步驟
   ├── 更新課程狀態為 COMPLETED
   ├── 將已出席學員狀態更新為 COMPLETED
   ├── 將未出席學員狀態更新為 NO_SHOW
   ├── 發布 CourseCompletedEvent
   └── 發布 TrainingCompletedEvent（每位完成學員）
```

Response Body

成功回應 (200 OK)

```
{
  "code": "SUCCESS",
  "message": "課程已結束",
  "data": {
    "courseId": "course-uuid-001",
    "status": "COMPLETED",
    "completedCount": 25,
    "noShowCount": 3,
    "completedAt": "2025-12-15T18:00:00Z"
  }
}
```

---

3. 報名管理 API

3.1 報名課程

基本資訊

項目	內容
端點	`POST /api/v1/training/enrollments`
方法	POST
Controller	HR10EnrollmentCmdController
Service	EnrollCourseServiceImpl
權限	`training:enrollment:create`

用途說明

項目	說明
業務場景	員工自行報名或 HR 代為報名課程
使用者	所有員工、HR 人員
後續動作	通知主管審核

業務邏輯

```
1. 驗證規則
   ├── 課程必須存在且狀態為 OPEN
   ├── 未超過報名截止日
   ├── 課程名額未滿
   ├── 員工未重複報名
   └── 員工符合先修條件（若有）

2. 處理步驟
   ├── 建立報名記錄，狀態為 REGISTERED
   ├── 發布 EnrollmentCreatedEvent
   └── Notification Service 通知主管審核

3. 自動核准規則
   ├── 免費內訓且非管理職 → 自動核准
   └── 其他 → 需主管審核
```

Request Body

```json
{
  "courseId": "course-uuid-001",
  "employeeId": "emp-uuid-001",
  "reason": "希望提升前端技術能力",
  "remarks": "已完成 React 基礎課程"
}
```

欄位	類型	必填	說明
`courseId`	UUID	✅	課程 ID
`employeeId`	UUID	❌	員工 ID（HR 代報時填寫，否則使用當前用戶）
`reason`	String	❌	報名理由
`remarks`	String	❌	備註

Response Body

成功回應 (201 Created)

```
{
  "code": "SUCCESS",
  "message": "報名成功，待主管審核",
  "data": {
    "enrollmentId": "enroll-uuid-001",
    "courseId": "course-uuid-001",
    "courseName": "React 進階開發實戰",
    "employeeId": "emp-uuid-001",
    "status": "REGISTERED",
    "enrolledAt": "2025-12-02T14:00:00Z"
  }
}
```

錯誤碼

HTTP 狀態碼	錯誤碼	說明	處理建議
400	`TRN_COURSE_NOT_OPEN`	課程未開放報名	請選擇開放中的課程
400	`TRN_ENROLLMENT_DEADLINE_PASSED`	報名截止	報名截止日已過
400	`TRN_COURSE_FULL`	課程已額滿	可申請候補
400	`TRN_ALREADY_ENROLLED`	已報名此課程	請勿重複報名

領域事件

EnrollmentCreatedEvent

```
{
  "eventType": "EnrollmentCreated",
  "aggregateId": "enroll-uuid-001",
  "aggregateType": "TrainingEnrollment",
  "payload": {
    "enrollmentId": "enroll-uuid-001",
    "courseId": "course-uuid-001",
    "courseName": "React 進階開發實戰",
    "employeeId": "emp-uuid-001",
    "employeeName": "張三",
    "managerId": "mgr-uuid-001",
    "managerName": "李組長",
    "trainingHours": 8,
    "cost": 0,
    "reason": "希望提升前端技術能力"
  }
}
```

Topic	訂閱服務
`training.enrollment.created`	Notification Service

3.2 查詢報名列表

基本資訊

項目	內容
端點	`GET /api/v1/training/enrollments`
方法	GET
Controller	HR10EnrollmentQryController
Service	GetEnrollmentsServiceImpl
權限	`training:enrollment:read`

Query Parameters

參數	類型	必填	說明
`courseId`	UUID	❌	課程篩選
`employeeId`	UUID	❌	員工篩選
`status`	Enum	❌	報名狀態
`pendingApproval`	Boolean	❌	待審核（主管視角）
`page`	Integer	❌	頁碼
`size`	Integer	❌	每頁筆數

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "data": {
    "items": [
      {
        "enrollmentId": "enroll-uuid-001",
        "course": {
          "courseId": "course-uuid-001",
          "courseCode": "TRN-202512-001",
          "courseName": "React 進階開發實戰",
          "startDate": "2025-12-15",
          "durationHours": 8
        },
        "employee": {
          "employeeId": "emp-uuid-001",
          "employeeNo": "EMP001",
          "employeeName": "張三",
          "departmentName": "研發部"
        },
        "status": "REGISTERED",
        "statusLabel": "待審核",
        "reason": "希望提升前端技術能力",
        "enrolledAt": "2025-12-02T14:00:00Z",
        "approvedAt": null,
        "approvedBy": null
      }
    ],
    "pagination": {
      "page": 1,
      "size": 20,
      "totalItems": 15,
      "totalPages": 1
    }
  }
}
```

3.3 審核通過

基本資訊

項目	內容
端點	`PUT /api/v1/training/enrollments/{id}/approve`
方法	PUT
Controller	HR10EnrollmentCmdController
Service	ApproveEnrollmentServiceImpl
權限	`training:enrollment:approve`

業務邏輯

```
1. 驗證規則
   ├── 報名必須存在
   ├── 報名狀態必須為 REGISTERED
   ├── 審核者必須是員工主管或 HR
   └── 課程名額未滿

2. 處理步驟
   ├── 更新報名狀態為 APPROVED
   ├── 記錄審核者與審核時間
   ├── 發布 EnrollmentApprovedEvent
   └── 通知員工報名成功
```

Request Body

```
{
  "remarks": "同意參訓"
}
```

Response Body

成功回應 (200 OK)

```
{
  "code": "SUCCESS",
  "message": "報名審核通過",
  "data": {
    "enrollmentId": "enroll-uuid-001",
    "status": "APPROVED",
    "approvedBy": "mgr-uuid-001",
    "approvedByName": "李組長",
    "approvedAt": "2025-12-03T09:00:00Z"
  }
}
```

領域事件

EnrollmentApprovedEvent

```
{
  "eventType": "EnrollmentApproved",
  "aggregateId": "enroll-uuid-001",
  "payload": {
    "enrollmentId": "enroll-uuid-001",
    "courseId": "course-uuid-001",
    "courseName": "React 進階開發實戰",
    "employeeId": "emp-uuid-001",
    "employeeName": "張三",
    "employeeEmail": "zhang@company.com",
    "startDate": "2025-12-15",
    "location": "台北辦公室 5F",
    "approvedBy": "mgr-uuid-001"
  }
}
```

Topic	訂閱服務
`training.enrollment.approved`	Notification Service

3.4 審核拒絕

基本資訊

項目	內容
端點	`PUT /api/v1/training/enrollments/{id}/reject`
方法	PUT
Controller	HR10EnrollmentCmdController
Service	RejectEnrollmentServiceImpl
權限	`training:enrollment:reject`

Request Body

```
{
  "reason": "目前專案進度緊迫，建議下期再報名"
}
```

欄位	類型	必填	說明
`reason`	String	✅	拒絕原因

Response Body

成功回應 (200 OK)

```
{
  "code": "SUCCESS",
  "message": "報名已拒絕",
  "data": {
    "enrollmentId": "enroll-uuid-001",
    "status": "REJECTED",
    "rejectedBy": "mgr-uuid-001",
    "rejectedAt": "2025-12-03T09:00:00Z"
  }
}
```

3.5 取消報名

基本資訊

項目	內容
端點	`PUT /api/v1/training/enrollments/{id}/cancel`
方法	PUT
Controller	HR10EnrollmentCmdController
Service	CancelEnrollmentServiceImpl
權限	`training:enrollment:cancel`

業務邏輯

```
1. 驗證規則
   ├── 報名必須存在
   ├── 報名狀態必須為 REGISTERED 或 APPROVED
   ├── 課程尚未開始
   └── 取消者必須是本人或 HR

2. 處理步驟
   ├── 更新報名狀態為 CANCELLED
   └── 發布 EnrollmentCancelledEvent
```

Request Body

```json
{
  "reason": "臨時有重要會議無法參加"
}
```

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "message": "報名已取消",
  "data": {
    "enrollmentId": "enroll-uuid-001",
    "status": "CANCELLED",
    "cancelledAt": "2025-12-05T10:00:00Z"
  }
}
```

3.6 確認出席

基本資訊

項目	內容
端點	`PUT /api/v1/training/enrollments/{id}/attendance`
方法	PUT
Controller	HR10EnrollmentCmdController
Service	ConfirmAttendanceServiceImpl
權限	`training:enrollment:attendance`

用途說明

項目	說明
業務場景	課程當天確認學員出席狀態
使用者	講師、HR 人員
可批次操作	支援批次確認多位學員出席

Request Body

```
{
  "attended": true,
  "attendedHours": 8,
  "remarks": "全程參與"
}
```

欄位	類型	必填	說明
`attended`	Boolean	✅	是否出席
`attendedHours`	Decimal	❌	實際出席時數（預設課程時數）
`remarks`	String	❌	備註

Response Body

成功回應 (200 OK)

```
{
  "code": "SUCCESS",
  "message": "出席已確認",
  "data": {
    "enrollmentId": "enroll-uuid-001",
    "status": "ATTENDED",
    "attendedHours": 8,
    "attendedAt": "2025-12-15T18:00:00Z"
  }
}
```

3.7 完成訓練

基本資訊

項目	內容
端點	`PUT /api/v1/training/enrollments/{id}/complete`
方法	PUT
Controller	HR10EnrollmentCmdController
Service	CompleteTrainingServiceImpl
權限	`training:enrollment:complete`

用途說明

項目	說明
業務場景	記錄學員完成訓練並登錄成績
使用者	講師、HR 人員
後續動作	更新員工訓練時數統計

業務邏輯

```
1. 驗證規則
   ├── 報名必須存在
   ├── 報名狀態必須為 ATTENDED
   └── 成績介於 0-100（若填寫）

2. 處理步驟
   ├── 更新報名狀態為 COMPLETED
   ├── 記錄成績與評語
   ├── 發布 TrainingCompletedEvent
   └── 更新員工訓練時數

3. 副作用
   └── Reporting Service 更新訓練統計
```

Request Body

```json
{
  "completedHours": 8,
  "score": 85,
  "passed": true,
  "feedback": "表現優異，課堂互動積極"
}
```

欄位	類型	必填	說明
`completedHours`	Decimal	❌	完成時數（預設出席時數）
`score`	Decimal	❌	成績（0-100）
`passed`	Boolean	❌	是否通過（預設 true）
`feedback`	String	❌	講師評語

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "message": "訓練已完成",
  "data": {
    "enrollmentId": "enroll-uuid-001",
    "status": "COMPLETED",
    "completedHours": 8,
    "score": 85,
    "passed": true,
    "completedAt": "2025-12-15T18:00:00Z"
  }
}
```

領域事件

TrainingCompletedEvent

```json
{
  "eventType": "TrainingCompleted",
  "aggregateId": "enroll-uuid-001",
  "aggregateType": "TrainingEnrollment",
  "payload": {
    "enrollmentId": "enroll-uuid-001",
    "employeeId": "emp-uuid-001",
    "employeeName": "張三",
    "courseId": "course-uuid-001",
    "courseName": "React 進階開發實戰",
    "courseCategory": "TECHNICAL",
    "completedHours": 8,
    "completedDate": "2025-12-15",
    "score": 85,
    "passed": true
  }
}
```

Topic	訂閱服務
`training.completed`	Reporting Service

3.8 我的訓練

基本資訊

項目	內容
端點	`GET /api/v1/training/my`
方法	GET
Controller	HR10EnrollmentQryController
Service	GetMyTrainingsServiceImpl
權限	`training:my:read`

用途說明

項目	說明
業務場景	員工查看自己的訓練記錄與進度
使用者	所有員工（ESS）
頁面代碼	HR10-P03

Query Parameters

參數	類型	必填	說明
`year`	Integer	❌	年度（預設當年）
`status`	Enum	❌	報名狀態
`category`	Enum	❌	課程類別

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "data": {
    "summary": {
      "year": 2025,
      "totalHours": 24,
      "requiredHours": 20,
      "completedCourses": 3,
      "upcomingCourses": 1,
      "isCompliant": true
    },
    "upcoming": [
      {
        "enrollmentId": "enroll-uuid-001",
        "courseId": "course-uuid-001",
        "courseName": "React 進階開發實戰",
        "startDate": "2025-12-15",
        "startTime": "09:00",
        "location": "台北辦公室 5F",
        "durationHours": 8,
        "status": "APPROVED"
      }
    ],
    "completed": [
      {
        "enrollmentId": "enroll-uuid-002",
        "courseId": "course-uuid-002",
        "courseName": "敏捷開發基礎",
        "completedDate": "2025-11-20",
        "completedHours": 8,
        "score": 90,
        "category": "MANAGEMENT"
      }
    ],
    "cancelled": [
      {
        "enrollmentId": "enroll-uuid-003",
        "courseName": "AWS 基礎課程",
        "cancelledDate": "2025-10-15",
        "reason": "課程取消"
      }
    ]
  }
}
```

4. 證照管理 API

4.1 新增證照

基本資訊

項目	內容
端點	`POST /api/v1/training/certificates`
方法	POST
Controller	HR10CertificateCmdController
Service	AddCertificateServiceImpl
權限	`training:certificate:create`

用途說明

項目	說明
業務場景	員工登錄取得的專業證照
使用者	所有員工、HR 人員
後續動作	到期前自動發送提醒

業務邏輯

```
1. 驗證規則
   ├── 證照名稱必填
   ├── 發證日期必填
   ├── 到期日（若有）需晚於發證日
   └── 附件大小限制 10MB

2. 處理步驟
   ├── 建立證照記錄
   ├── 上傳附件至 Document Service
   └── 發布 CertificateAddedEvent

3. 到期提醒規則
   ├── 到期前 90 天：第一次提醒
   ├── 到期前 30 天：第二次提醒
   └── 到期前 7 天：緊急提醒
```

Request Body

```
{
  "employeeId": "emp-uuid-001",
  "certificateName": "AWS Solutions Architect - Associate",
  "issuingOrganization": "Amazon Web Services",
  "certificateNumber": "AWS-SAA-123456",
  "issueDate": "2023-12-15",
  "expiryDate": "2026-12-15",
  "category": "TECHNICAL",
  "isRequired": true,
  "attachmentUrl": "https://storage.company.com/certs/aws-saa.pdf",
  "remarks": "公司補助考照費用"
}
```

欄位	類型	必填	說明
`employeeId`	UUID	❌	員工 ID（預設當前用戶）
`certificateName`	String	✅	證照名稱
`issuingOrganization`	String	❌	發證機構
`certificateNumber`	String	❌	證照編號
`issueDate`	Date	✅	發證日期
`expiryDate`	Date	❌	到期日期（永久有效則不填）
`category`	Enum	❌	證照類別
`isRequired`	Boolean	❌	是否為必備證照
`attachmentUrl`	String	❌	證照附件
`remarks`	String	❌	備註

Response Body

成功回應 (201 Created)

```json
{
  "code": "SUCCESS",
  "message": "證照新增成功",
  "data": {
    "certificateId": "cert-uuid-001",
    "certificateName": "AWS Solutions Architect - Associate",
    "employeeId": "emp-uuid-001",
    "issueDate": "2023-12-15",
    "expiryDate": "2026-12-15",
    "daysUntilExpiry": 350,
    "isVerified": false,
    "createdAt": "2025-12-30T10:00:00Z"
  }
}
```

4.2 查詢證照列表

基本資訊

項目	內容
端點	`GET /api/v1/training/certificates`
方法	GET
Controller	HR10CertificateQryController
Service	GetCertificatesServiceImpl
權限	`training:certificate:read`

Query Parameters

參數	類型	必填	說明
`employeeId`	UUID	❌	員工篩選
`category`	Enum	❌	類別篩選
`isExpired`	Boolean	❌	是否已過期
`isRequired`	Boolean	❌	是否必備
`keyword`	String	❌	關鍵字搜尋
`page`	Integer	❌	頁碼
`size`	Integer	❌	每頁筆數

Response Body

成功回應 (200 OK)

```
{
  "code": "SUCCESS",
  "data": {
    "items": [
      {
        "certificateId": "cert-uuid-001",
        "employee": {
          "employeeId": "emp-uuid-001",
          "employeeNo": "EMP001",
          "employeeName": "張三",
          "departmentName": "研發部"
        },
        "certificateName": "AWS Solutions Architect - Associate",
        "issuingOrganization": "Amazon Web Services",
        "certificateNumber": "AWS-SAA-123456",
        "issueDate": "2023-12-15",
        "expiryDate": "2026-12-15",
        "daysUntilExpiry": 350,
        "status": "VALID",
        "statusLabel": "有效",
        "category": "TECHNICAL",
        "categoryLabel": "技術類",
        "isRequired": true,
        "isVerified": true,
        "attachmentUrl": "https://storage.company.com/certs/aws-saa.pdf"
      }
    ],
    "summary": {
      "total": 45,
      "valid": 40,
      "expiring": 3,
      "expired": 2
    },
    "pagination": {
      "page": 1,
      "size": 20,
      "totalItems": 45,
      "totalPages": 3
    }
  }
}
```

4.3 查詢證照詳情

基本資訊

項目	內容
端點	`GET /api/v1/training/certificates/{id}`
方法	GET
Controller	HR10CertificateQryController
Service	GetCertificateDetailServiceImpl
權限	`training:certificate:read`

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "data": {
    "certificateId": "cert-uuid-001",
    "employee": {
      "employeeId": "emp-uuid-001",
      "employeeNo": "EMP001",
      "employeeName": "張三",
      "departmentName": "研發部"
    },
    "certificateName": "AWS Solutions Architect - Associate",
    "issuingOrganization": "Amazon Web Services",
    "certificateNumber": "AWS-SAA-123456",
    "issueDate": "2023-12-15",
    "expiryDate": "2026-12-15",
    "daysUntilExpiry": 350,
    "status": "VALID",
    "category": "TECHNICAL",
    "isRequired": true,
    "isVerified": true,
    "verifiedBy": "HR Admin",
    "verifiedAt": "2024-01-10T10:00:00Z",
    "attachmentUrl": "https://storage.company.com/certs/aws-saa.pdf",
    "remarks": "公司補助考照費用",
    "history": [
      {
        "date": "2023-12-15",
        "action": "CREATED",
        "description": "證照建立"
      },
      {
        "date": "2024-01-10",
        "action": "VERIFIED",
        "description": "HR 驗證通過"
      }
    ],
    "createdAt": "2023-12-20T10:00:00Z",
    "updatedAt": "2024-01-10T10:00:00Z"
  }
}
```

4.4 更新證照

基本資訊

項目	內容
端點	`PUT /api/v1/training/certificates/{id}`
方法	PUT
Controller	HR10CertificateCmdController
Service	UpdateCertificateServiceImpl
權限	`training:certificate:update`

Request Body

```
{
  "certificateNumber": "AWS-SAA-123456-RENEWED",
  "issueDate": "2025-12-15",
  "expiryDate": "2028-12-15",
  "attachmentUrl": "https://storage.company.com/certs/aws-saa-renewed.pdf",
  "remarks": "已續證"
}
```

Response Body

成功回應 (200 OK)

```
{
  "code": "SUCCESS",
  "message": "證照更新成功",
  "data": {
    "certificateId": "cert-uuid-001",
    "certificateName": "AWS Solutions Architect - Associate",
    "expiryDate": "2028-12-15",
    "updatedAt": "2025-12-30T10:00:00Z"
  }
}
```

4.5 刪除證照

基本資訊

項目	內容
端點	`DELETE /api/v1/training/certificates/{id}`
方法	DELETE
Controller	HR10CertificateCmdController
Service	DeleteCertificateServiceImpl
權限	`training:certificate:delete`

業務邏輯

```
1. 驗證規則
   ├── 證照必須存在
   ├── 刪除者必須是本人或 HR
   └── 必備證照需 HR 審核後才能刪除

2. 處理步驟
   ├── 軟刪除證照記錄
   └── 發布 CertificateDeletedEvent
```

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "message": "證照已刪除"
}
```

4.6 即將到期證照

基本資訊

項目	內容
端點	`GET /api/v1/training/certificates/expiring`
方法	GET
Controller	HR10CertificateQryController
Service	GetExpiringCertificatesServiceImpl
權限	`training:certificate:read`

用途說明

項目	說明
業務場景	HR 監控即將到期的證照，提醒員工續證
使用者	HR 人員、主管
排程任務	每日自動檢查並發送提醒

Query Parameters

參數	類型	必填	說明
`days`	Integer	❌	到期天數內（預設 30）
`departmentId`	UUID	❌	部門篩選
`isRequired`	Boolean	❌	僅顯示必備證照

Response Body

成功回應 (200 OK)

```
{
  "code": "SUCCESS",
  "data": {
    "items": [
      {
        "certificateId": "cert-uuid-002",
        "employee": {
          "employeeId": "emp-uuid-002",
          "employeeName": "李四",
          "email": "li@company.com",
          "departmentName": "研發部"
        },
        "certificateName": "PMP 專案管理師",
        "issuingOrganization": "PMI",
        "expiryDate": "2026-01-15",
        "daysUntilExpiry": 16,
        "isRequired": true,
        "urgencyLevel": "HIGH"
      }
    ],
    "summary": {
      "within7Days": 2,
      "within30Days": 5,
      "within90Days": 8
    }
  }
}
```

領域事件

CertificateExpiringEvent（排程任務發布）

```
{
  "eventType": "CertificateExpiring",
  "aggregateId": "cert-uuid-002",
  "aggregateType": "Certificate",
  "payload": {
    "certificateId": "cert-uuid-002",
    "employeeId": "emp-uuid-002",
    "employeeName": "李四",
    "employeeEmail": "li@company.com",
    "certificateName": "PMP 專案管理師",
    "issuingOrganization": "PMI",
    "expiryDate": "2026-01-15",
    "daysUntilExpiry": 16,
    "isRequired": true
  }
}
```

Topic	訂閱服務
`training.certificate.expiring`	Notification Service

5. 訓練統計 API

5.1 我的訓練時數

基本資訊

項目	內容
端點	`GET /api/v1/training/my/hours`
方法	GET
Controller	HR10ReportQryController
Service	GetMyTrainingHoursServiceImpl
權限	`training:my:read`

Query Parameters

參數	類型	必填	說明
`year`	Integer	❌	年度（預設當年）

Response Body

成功回應 (200 OK)

```json
{
  "code": "SUCCESS",
  "data": {
    "year": 2025,
    "employee": {
      "employeeId": "emp-uuid-001",
      "employeeName": "張三",
      "departmentName": "研發部"
    },
    "summary": {
      "totalHours": 24,
      "requiredHours": 20,
      "remainingHours": 0,
      "isCompliant": true,
      "complianceRate": 120
    },
    "byCategory": [
      {
        "category": "TECHNICAL",
        "categoryLabel": "技術類",
        "hours": 16,
        "courseCount": 2
      },
      {
        "category": "MANAGEMENT",
        "categoryLabel": "管理類",
        "hours": 8,
        "courseCount": 1
      }
    ],
    "byMonth": [
      {"month": "2025-01", "hours": 0},
      {"month": "2025-02", "hours": 0},
      {"month": "2025-03", "hours": 8},
      {"month": "2025-04", "hours": 0},
      {"month": "2025-05", "hours": 0},
      {"month": "2025-06", "hours": 0},
      {"month": "2025-07", "hours": 8},
      {"month": "2025-08", "hours": 0},
      {"month": "2025-09", "hours": 0},
      {"month": "2025-10", "hours": 0},
      {"month": "2025-11", "hours": 8},
      {"month": "2025-12", "hours": 0}
    ],
    "recentTrainings": [
      {
        "courseName": "React 進階開發實戰",
        "completedDate": "2025-11-20",
        "hours": 8,
        "category": "TECHNICAL"
      }
```

```
      ]
    }
  }
```

5.2 訓練統計報表

基本資訊

項目	內容
端點	`GET /api/v1/training/statistics`
方法	GET
Controller	HR10ReportQryController
Service	GetTrainingStatisticsServiceImpl
權限	`training:report:read`

用途說明

項目	說明
業務場景	HR 主管查看整體訓練成效
使用者	HR 主管、人資長
頁面代碼	HR10-P06

Query Parameters

參數	類型	必填	說明
`year`	Integer	❌	年度（預設當年）
`departmentId`	UUID	❌	部門篩選

Response Body

成功回應 (200 OK)

```
{
  "code": "SUCCESS",
  "data": {
    "year": 2025,
    "overview": {
      "totalCourses": 45,
      "completedCourses": 38,
      "totalEnrollments": 520,
      "totalCompletions": 485,
      "completionRate": 93.3,
      "totalTrainingHours": 3200,
      "totalBudget": 500000,
      "usedBudget": 380000,
      "budgetUtilization": 76
    },
    "complianceStatus": {
      "totalEmployees": 150,
      "compliantEmployees": 142,
      "complianceRate": 94.7,
      "nonCompliantList": [
        {
          "employeeId": "emp-uuid-003",
          "employeeName": "王五",
          "departmentName": "業務部",
          "completedHours": 12,
          "requiredHours": 20,
          "shortfall": 8
        }
      ]
    },
    "byDepartment": [
      {
        "departmentId": "dept-uuid-001",
        "departmentName": "研發部",
        "employeeCount": 50,
        "totalHours": 1200,
        "avgHoursPerEmployee": 24,
        "complianceRate": 98
      }
    ],
    "byCategory": [
      {
        "category": "TECHNICAL",
        "categoryLabel": "技術類",
        "courseCount": 20,
        "totalHours": 1500,
        "participantCount": 280
      }
    ],
    "topCourses": [
      {
```

```
        "courseId": "course-uuid-001",
        "courseName": "React 進階開發實戰",
        "enrollmentCount": 45,
        "completionRate": 95,
        "avgScore": 88
      }
    ],
    "monthlyTrend": [
      {"month": "2025-01", "courses": 3, "hours": 240},
      {"month": "2025-02", "courses": 4, "hours": 320}
    ]
  }
}
```

5.3 匯出統計報表

基本資訊

項目	內容
端點	`GET /api/v1/training/statistics/export`
方法	GET
Controller	HR10ReportQryController
Service	ExportTrainingStatisticsServiceImpl
權限	`training:report:export`

Query Parameters

參數	類型	必填	說明
`year`	Integer	❌	年度
`departmentId`	UUID	❌	部門篩選
`format`	Enum	❌	格式：EXCEL / PDF（預設 EXCEL）
`reportType`	Enum	❌	SUMMARY / DETAIL / COMPLIANCE

Response

成功回應 (200 OK)

```
Content-Type: application/vnd.openxmlformats-officedocument.spreadsheetml.sheet
Content-Disposition: attachment; filename="training_report_2025.xlsx"
```

---

6. 錯誤碼定義

6.1 錯誤碼列表

錯誤碼	HTTP 狀態碼	說明	處理建議
課程相關			
`TRN_COURSE_NOT_FOUND`	404	課程不存在	請確認課程 ID
`TRN_COURSE_NOT_OPEN`	400	課程未開放報名	請選擇開放中的課程
`TRN_COURSE_NOT_EDITABLE`	400	課程不可編輯	只有草稿或開放狀態可編輯
`TRN_COURSE_NAME_REQUIRED`	400	課程名稱必填	請填寫課程名稱
`TRN_COURSE_CODE_EXISTS`	400	課程代碼已存在	請使用其他代碼
`TRN_INVALID_DATE_RANGE`	400	日期範圍無效	結束日期需 >= 開始日期
`TRN_INVALID_HOURS`	400	訓練時數無效	時數需 > 0
`TRN_COURSE_FULL`	400	課程已額滿	可申請候補
報名相關			
`TRN_ENROLLMENT_NOT_FOUND`	404	報名不存在	請確認報名 ID
`TRN_ENROLLMENT_DEADLINE_PASSED`	400	報名截止	報名截止日已過
`TRN_ALREADY_ENROLLED`	400	已報名此課程	請勿重複報名
`TRN_INVALID_ENROLLMENT_STATUS`	400	報名狀態不符	無法執行此操作
`TRN_NOT_APPROVER`	403	非審核者	您無權審核此報名
`TRN_CANCEL_NOT_ALLOWED`	400	無法取消報名	課程已開始
證照相關			
`TRN_CERTIFICATE_NOT_FOUND`	404	證照不存在	請確認證照 ID

錯誤碼	HTTP 狀態碼	說明	處理建議
`TRN_CERTIFICATE_NAME_REQUIRED`	400	證照名稱必填	請填寫證照名稱
`TRN_INVALID_EXPIRY_DATE`	400	到期日無效	到期日需晚於發證日
`TRN_DELETE_REQUIRED_CERT`	400	無法刪除必備證照	請聯繫 HR
權限相關			
`TRN_NO_PERMISSION`	403	無權限	請聯繫管理員

7. 領域事件

7.1 事件清單

事件名稱	Kafka Topic	觸發時機	訂閱服務
`CourseCreated`	`training.course.created`	建立課程	-
`CoursePublished`	`training.course.published`	發布課程	Notification（必修通知）
`CourseClosed`	`training.course.closed`	關閉報名	-
`CourseCompleted`	`training.course.completed`	課程結束	-
`EnrollmentCreated`	`training.enrollment.created`	報名課程	Notification
`EnrollmentApproved`	`training.enrollment.approved`	審核通過	Notification
`EnrollmentRejected`	`training.enrollment.rejected`	審核拒絕	Notification
`EnrollmentCancelled`	`training.enrollment.cancelled`	取消報名	-
`TrainingCompleted`	`training.completed`	完成訓練	Reporting
`CertificateAdded`	`training.certificate.added`	新增證照	-
`CertificateExpiring`	`training.certificate.expiring`	證照即將到期	Notification

7.2 證照到期檢查排程

```
排程任務: CertificateExpiryCheckScheduler
執行頻率: 每日 08:00
處理邏輯:
  1. 查詢 90 天內到期的證照
  2. 依據到期天數分類（7/30/90 天）
  3. 發布 CertificateExpiringEvent
  4. Notification Service 發送提醒郵件
```

文件完成日期: 2025-12-30 **API 端點總數:** 24 個